



From Monocular Visual Odometry to Visual-Inertial Odometry on TUM VI

Presented By: **Fiyinfoluwa Ajepe**
Jordan Fogang

May 2026 1

Background

- Autonomous robots (drones, AGVs, warehouse logistics) require precise, continuous ego-motion estimation to navigate safely.
- Traditional satellite positioning completely degrades or fails inside buildings, tunnels, and deep urban canyons.
- VO Serves as a lightweight, camera-driven solution, incrementally tracking position and orientation using purely geometric frame-to-frame changes.

The Monocular Limitation (Scale Ambiguity)

- A single camera can easily capture directional shifts, but it is physically incapable of determining absolute real-world size or metric depth
- In epipolar geometry, the Essential Matrix (E) maps relative motion, but the translation vector (t) is strictly constrained to an arbitrary unit length

$$\mathbf{x}^T \mathbf{E} \mathbf{x} = 0, \text{ where } \mathbf{E} = [\mathbf{t}] \times \mathbf{R}, \|\mathbf{t}\| = 1$$

- To a flat image sensor, a camera moving 1 meter through a standard room looks identical to a camera moving 100 meter through a giant stadium. This lack of scale leads to massive trajectory drift over time.

The VIO Solution

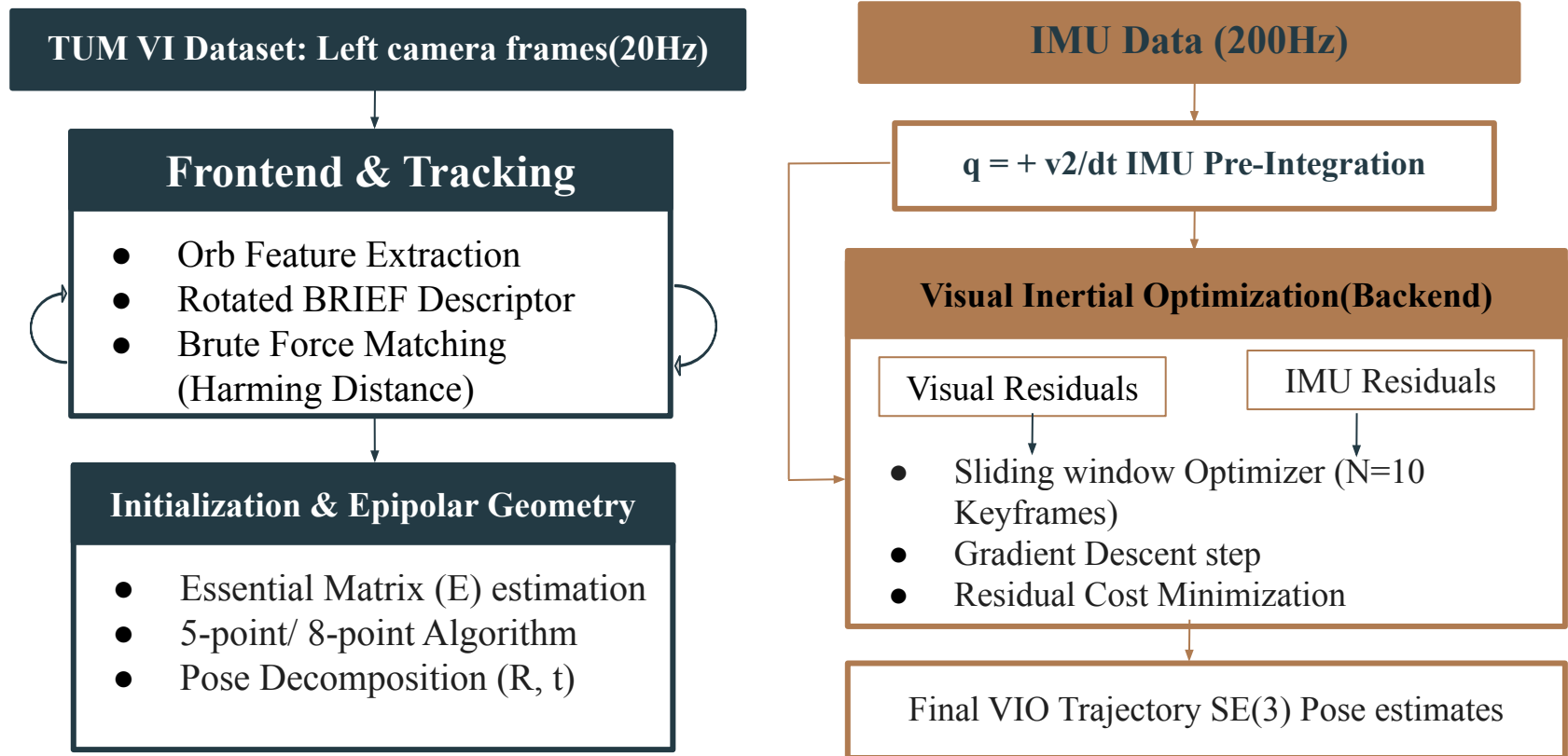
- Fusing camera frames with an Inertial Measurement Unit (IMU) injects physical metric constraints (m/s) into our system via raw acceleration measurements.
- The IMU operates at a much higher frequency (200Hz) than the camera (20Hz), providing continuous state estimates between video frames.
- Camera: Ideal for slow, precise, and visually rich tracking.
- IMU: Ideal during sudden, aggressive maneuvers or temporary motion blur where feature tracking slips.

Objectives

The Objectives of this paper is to:

- Develop a classical, feature-based Monocular Visual Odometry (VO) baseline from scratch.
- Integrate high-frequency IMU data through manifold space pre-integration.
- Build a tightly-coupled sliding-window optimizer to fuse visual and inertial residuals.
- Conduct a rigorous performance analysis using ATE and RPE metrics on the TUM VI dataset.

Pipeline Overview



Methodology- Feature Detection & Tracking

- ORB was used to detect and describe the distinctive keypoints efficiently in real time.
- Keypoints were matched across consecutive frames using Brute-Force Hamming Distance.
- Two-Step Outlier Rejection :

Lowe's Ratio Test: This sets a strict threshold ($\tau = 0.75$) to filter out ambiguous, repetitive descriptors.

Cross-Matching Check: This enforces bidirectional matching consistency to eliminate false positives.

Methodology- Initialization & Epipolar Geometry

- Establishes the initial 3D world coordinate frame using the first two consecutive keyframes.
- The Essential Matrix (E) was computed from tracked pixel correspondences using the 8-Point Algorithm inside a robust RANSAC loop.
- Decomposes (E) into a unique Relative Rotation (R) and Translation (t) via Singular Value Decomposition (SVD), enforcing the chirality constraint
- Because this is pure monocular vision at this stage, the initial translation vector is normalized to unit length ($\|t\| = 1$), fixing the arbitrary baseline scale.

Methodology- VIO Backend & IMU Preintegration

- Integrates high-frequency inertial measurements (200Hz) between visual frames in a local manifold space, preventing the need to re-integrate states during optimization.
- A fixed-lag window of the last $N = 10$ keyframes was maintained to keep computation bounded while enforcing historical motion history.
- Formulates a combined non-linear cost function to minimize both visual and inertial errors simultaneously
- Utilizes a simplified gradient-descent step to iteratively minimize joint residuals across the active window

Experimental Setup

- Evaluated thoroughly on the TUM VI Benchmark Dataset, utilizing high-resolution, time-synchronized monocular camera frames (20Hz) and IMU data (200Hz)
 - ❖ Room2 (Office): Full continuous ground truth; standard environment for baseline verification.
 - ❖ Corridor3: Structural symmetry, geometric self-similarity, and low-texture sequences; features start-end ground truth.
 - ❖ Outdoors5: Long-range loops with lighting variations and massive scale drift potential.

Experimental Setup

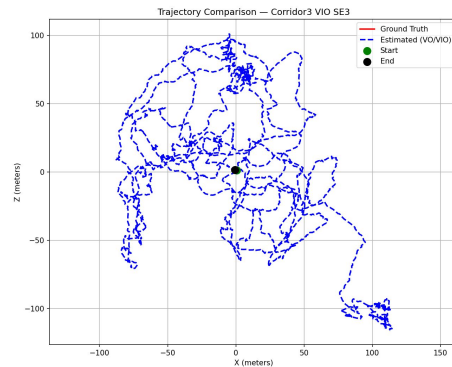
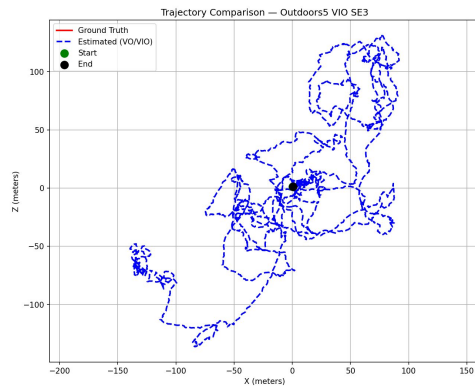
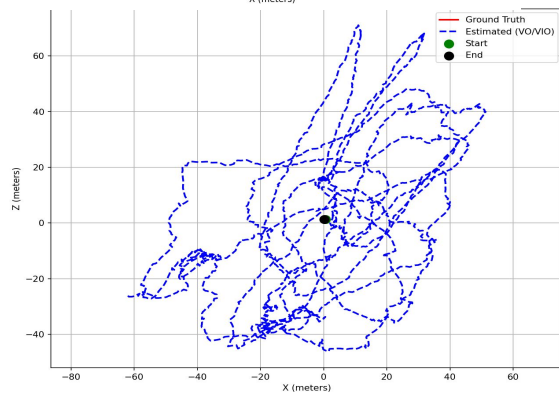
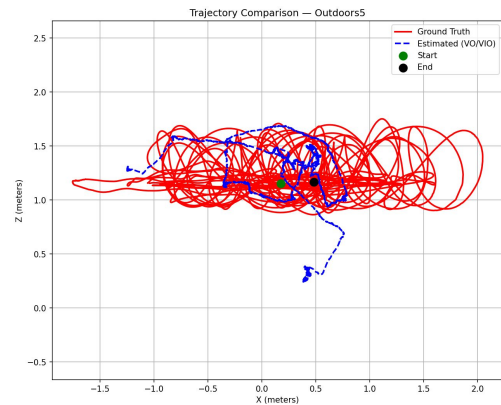
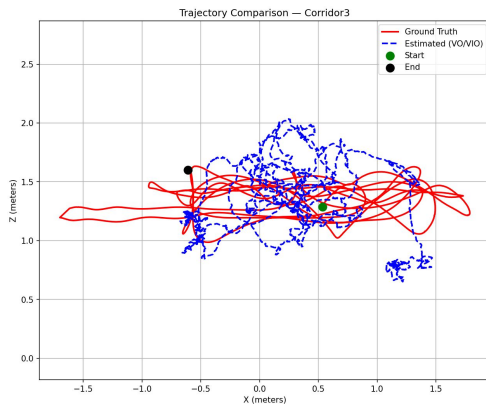
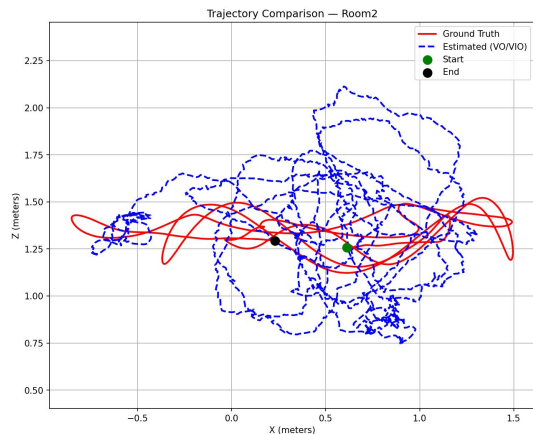
- Evaluated using Absolute Trajectory Error (ATE) and Relative Pose Error (RPE) to separate local tracking precision from global drift.
- Standalone VO is evaluated via Sim3 alignment (accounting for scale), while VIO is evaluated under strict SE3 rigid alignment to explicitly test metric scale recovery.

Results

Sequence	ATE (m)	RPE (m/100m)
Room2	0.7224	0.0150
Corridor3	1.1316	0.0107
Outdoors5	1.0876	0.0107

- The exceptionally low Relative Pose Error (**RPE $\approx$ 0.01m/100m**) proves that our frontend tracking and epipolar geometry are highly precise frame-to-frame.
- In standalone VO, scale ambiguity causes severe unconstrained drift over time, reaching **321.81m** on Corridor3 and **978.22m** on Outdoors5.
- Evaluating our VIO pipeline under rigid **SE(3) alignment** reveals that metric scale did not fully converge (resulting in a Room2 ATE of 0.7564m).

Results



Future Works & Conclusion

Future Works

- We plan on moving from our first-order gradient descent step to a second-order batch optimizer (such as **g2o**) for more stable and precise state convergence.

Conclusion

Developed a feature-based Monocular VO baseline and successfully transitioned to a tightly-coupled VIO framework. We also demonstrated that the fundamental scale ambiguity of monocular vision can be effectively anchored using physical inertial measurements.